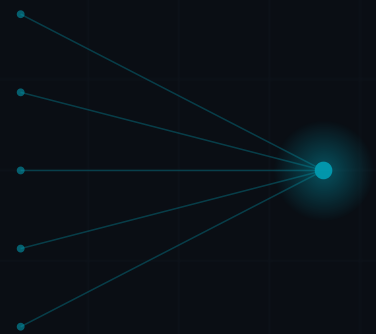


INDUSTRIAL SAFETY INTELLIGENCE

CORRIX



The correlation layer industrial safety never had.

An AI-powered platform that fuses gas sensors, work permits, shift schedules, worker location, and computer vision into one reasoning layer, and catches the compound risk that no single system is watching.

PROBLEM STATEMENT 1

AI-Powered Industrial Safety Intelligence for Zero-Harm Operations

TEAM

Hack4Impact!

Sumantha · Shreyas S · Syed Ayaan · Sharan Malali

Team and executive summary

The team

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What we built, in one paragraph

Corrix is an industrial safety intelligence platform built for Problem Statement 1. It fuses five normally-siloed data streams into one reasoning layer that detects compound risk: dangerous combinations of individually-ordinary conditions that no single sensor, system, or team would flag alone. When a compound risk is detected, a five-agent AI Safety Council convenes, reaches an explainable verdict grounded in cited regulation, forecasts time-to-critical as a probability band, computes an evacuation route that avoids other unsafe zones, and can fire a real emergency notification carrying a hashed evidence snapshot. Everything surfaces on a live command-center dashboard, and a second front door lets any facility onboard its own zones and stream live data over MQTT into the same engine.

Why it stands out

- 01 The core thesis is enforced by architecture, not asserted.** Each Council agent can only query the data source its real-world counterpart would have. No single agent can see the whole picture, so compound risk genuinely emerges from fusion.
- 02 It catches what nobody scripted.** A joint-evidence novelty detector escalates statistically unusual combinations that match no known pattern, so the system is not a memorized list of demo scenarios.
- 03 Every claim is measured.** An evaluation harness scores the pipeline against a labeled scenario library with matched negative controls, including a held-out split that proves the learning loop generalizes.
- 04 It is radically honest.** Every real subsystem is labeled real, every simulated stream is labeled simulated, on screen, inside the product. The simulator itself is validated against a real industrial dataset.
- 05 It is built, tested, and deployed.** 415 passing backend tests and a live public instance.

The problem the brief describes

Problem Statement 1 opens with a hard fact about Indian heavy industry: the sector continues to pay a devastating human cost. The brief cites over 6,500 fatal workplace accidents recorded in FY2023 according to DGFASLI, a figure that excludes most of the mining and construction sectors. Independent public reporting corroborates the scale: India records roughly one serious industrial accident every two days in registered factories.

The brief's own diagnosis: "data present, but unacted upon." The problem is not the absence of technology. It is the absence of a unified intelligence layer that fuses data from disparate sensors, shift logs, maintenance records, and video feeds into a real-time risk picture, and acts on it before a fatality, not after.

The brief also cites a 2024 FICCI survey finding that over 60 percent of large industrial facilities rely on manual handoffs to coordinate between their own digital safety tools. Plants are not under-instrumented. They are under-connected. Each safety system is individually competent and collectively blind to the others:

Gas / IoT sensors See: gas concentration per zone Blind: active permits nearby	SCADA See: process parameters Blind: permits, people in zone	Permit-to-work See: who is authorized, where Blind: live sensor conditions	CCTV See: visual activity Blind: readings, permit status	Shift logs See: historical work record Blind: real-time correlation
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The challenge statement asks for a single predictive layer over IoT sensors, SCADA, permit-to-work logs, CCTV feeds, and shift records, one that detects compound risk conditions no single sensor would flag alone and triggers preemptive interventions before they escalate. That sentence is the entire design brief for Corrix. We did not treat the six illustrative sub-directions as six separate features to check off. We treated the underlying failure mode as one problem and built one system around it.

The anchor incident, cited accurately

The brief grounds the problem in a real tragedy at the Visakhapatnam Steel Plant. We took that anchor seriously enough to verify it, and what we found shaped the project's character.

On 8 June 2025, eight workers at the Visakhapatnam Steel Plant were killed when entrapped gases inside a ladle of molten steel triggered a sudden explosion during a routine casting operation at Steel Melting Shop 2. A preliminary investigation by the Chief Inspector of Factories, Andhra Pradesh, attributed the explosion to gas entrapped in the liquid steel. The event is confirmed across multiple independent outlets, including The Wire, whose headline matches the brief's own phrasing about entrapped gases.

8

WORKERS LOST

5

SAFETY SYSTEMS RUNNING

0

SYSTEMS CORRELATING THEM

The brief describes a January 2025 coke-oven battery incident at the same plant. Our research could not independently verify that specific framing; the closest verified match in public reporting is the June 2025 ladle event described above. Corrix cites the version that checks out, and we state the correction openly rather than repeating a detail we could not source.

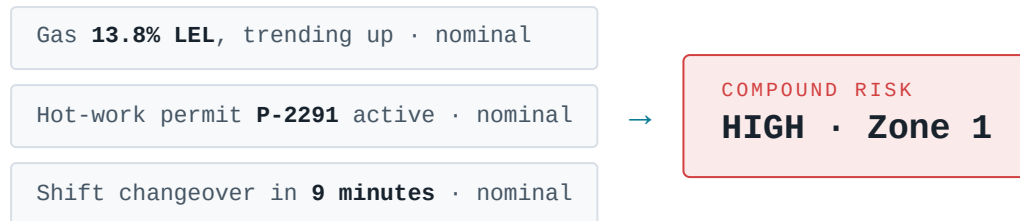
Why this matters: the correction does not weaken the thesis, it sharpens it. The signal, gas entrapment in an in-process vessel, existed in the physical system. No layer connected that condition to the fact that a lifting and casting operation was actively underway in that exact location, at that exact moment. People died in the gap between systems that each held one piece of the answer. That gap is precisely what Corrix closes.

The pattern behind the incident

Read as a systems failure rather than a single malfunction, the incident has a structure: a physical process condition, an authorized human operation, a location, and a moment in time, each tracked by a different system, none of them cross-referenced. This structure repeats across the compound-risk examples the brief itself names: maintenance activity co-occurring with gas accumulation, confined space entry during abnormal process conditions, hot work permits near elevated gas readings. Corrix is built to detect this structure in general, not any one instance of it.

The insight: compound risk is a reasoning problem

Every condition in a compound risk is individually routine. That is exactly why threshold-based systems miss it.



No alarm fires on any of the three inputs, because none of them crosses a limit. The danger lives in their intersection, in one zone, at one moment. Detecting it requires holding all the facts at once and exercising judgment about their combination. That is a cross-source reasoning problem, and it dictated our architecture. We deliberately ruled out three simpler shapes:

SIMPLER APPROACH	WHY IT FAILS THIS PROBLEM
One anomaly model over all sensors	Treats a reasoning problem as pattern recognition in one signal. Needs training volume that does not exist, is hard to explain to a safety officer, and cannot articulate why a combination is dangerous.
A computer-vision safety platform	The commercial incumbents are behavior-based safety cameras: PPE compliance, posture, vehicle proximity. None fuse permit state, process trend, and shift context into a predictive layer for process-safety incidents such as gas entrapment. The Vizag-class incident is a process-safety failure, not a behavior-safety one.
A rule engine alone	Deterministic rules are essential for the auditable core, and Corrix keeps them. But rules cannot express the judgment a multi-disciplinary safety review makes: none of these three things alone is a problem, but together, right now, they are.

The design that follows is a hybrid: a fast, deterministic, fully-auditable statistical layer for detection, and a multi-agent reasoning layer for judgment, mirroring how real incident reviews actually work. A process engineer, a permit officer, and an operations lead each examine the same event through a different lens, and their convergence or disagreement is itself the signal. Corrix reproduces that structure with AI agents, and makes each agent's reasoning visible, so a safety officer can audit the judgment per perspective rather than trusting a single opaque score.

What Corrix does

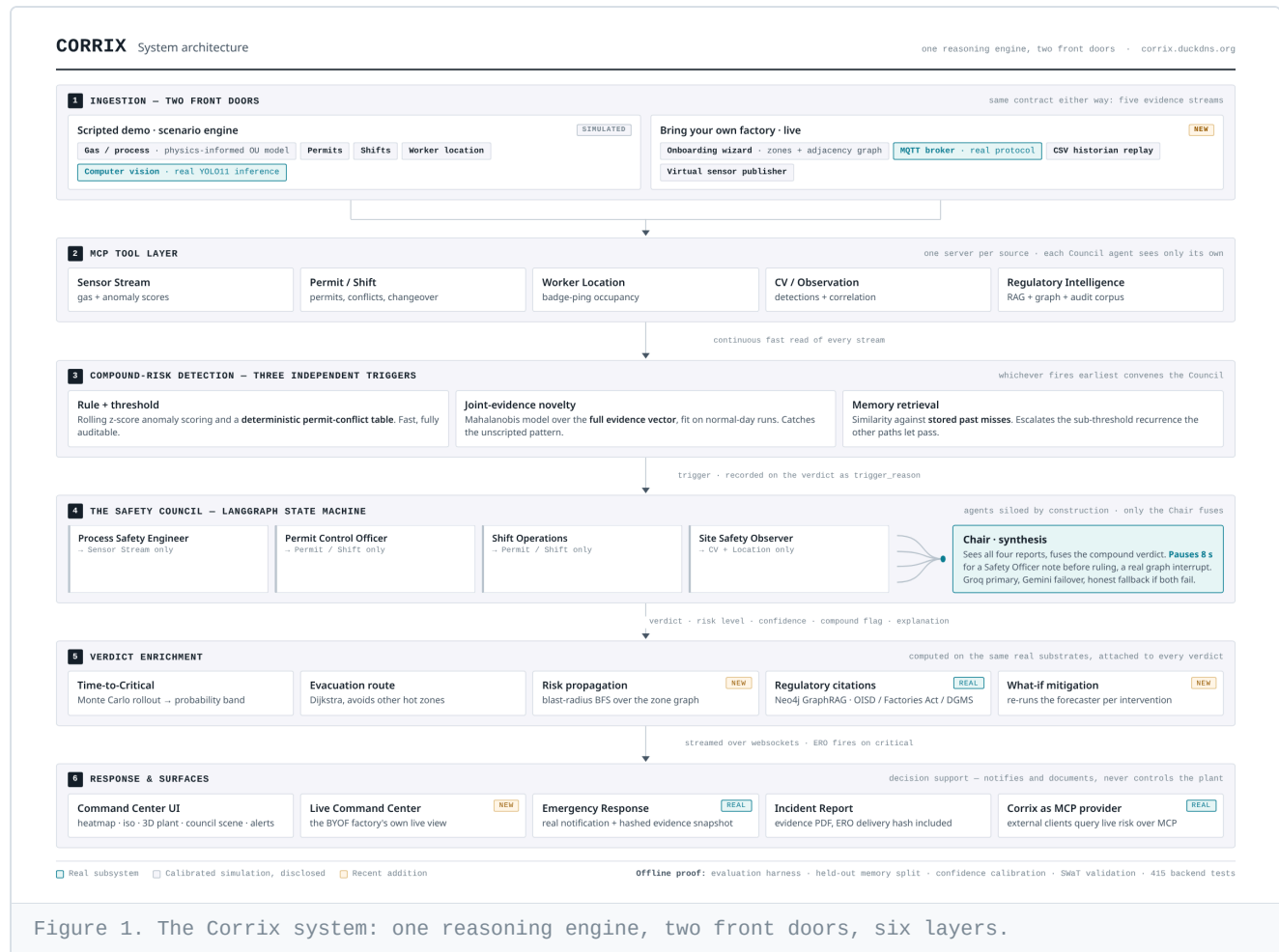
Corrix covers all six of the brief's illustrative sub-directions, deliberately consolidated into four integrated pillars rather than six disconnected modules. A system that visibly synthesizes the brief into a coherent whole serves a safety officer better than six panels competing for attention.

PILLAR	COVERS THE BRIEF'S	WHAT IT ACTUALLY IS
1. Compound Risk Detection	Compound Risk Detection Engine; Digital Permit Intelligence Agent	Three independent detection triggers feeding a five-agent Safety Council. Permit intelligence is a first-class evidence stream inside the same engine, not a separate bolt-on.
2. Geospatial Situational Awareness	Geospatial Safety Heatmap	A live plant heatmap with worker-location markers, risk-aware evacuation routing, spatial risk propagation, and flat, isometric, and 3D views. One map, several live layers.
3. Regulatory Intelligence	Incident Pattern Intelligence; Quality & Compliance Audit Agent	One Neo4j GraphRAG substrate over real OISD, Factories Act 1948, and DGMS text serving three query shapes: regulation Q&A, historical pattern lookup, and compliance checks. Every verdict carries its citation.
4. Emergency Response	Emergency Response Orchestrator	On a CRITICAL verdict: a real notification with a timestamped, SHA-256-hashed evidence snapshot and the computed evacuation route, plus a one-click PDF incident report.

Beyond the brief

- **Time-to-Critical forecasting:** a Monte Carlo rollout produces a per-zone probability band, honest uncertainty instead of a falsely-precise countdown.
- **What-if mitigation:** a safety officer can test an intervention and see the predicted change in time-to-critical before acting.
- **A self-improving memory loop:** the system learns from its own past misses, with the improvement measured only on held-out cases.
- **Safety Officer Override:** a real human-in-the-loop interrupt before any verdict is issued.
- **Bring Your Own Factory:** onboard any facility and stream live data over MQTT into the same engine.
- **Dual-role MCP integration:** Corrix consumes its data sources over the Model Context Protocol and exposes its own risk state back out over the same protocol.

Architecture



- 01 Ingestion, two front doors.** A scripted scenario engine generates physics-informed synthetic streams for the demo; the live Bring Your Own Factory path accepts real MQTT, CSV historian replay, or virtual sensors. Both produce the same five evidence streams.
- 02 MCP tool layer.** Every data source is a Model Context Protocol server. Each Council agent is scoped to exactly one, which is what makes the silo design enforceable.
- 03 Compound-risk detection.** Three independent triggers watch every stream continuously; any one can convene the Council, and the verdict records which fired.
- 04 The Safety Council.** A LangGraph state machine: four siloed evidence agents and a synthesizing Chair, with a real 8-second human-override interrupt before the Chair rules.
- 05 Verdict enrichment.** Time-to-critical, evacuation route, risk propagation, what-if analysis, and regulatory citations are computed on the same substrates and attached to every verdict.
- 06 Response and surfaces.** The command-center dashboard, the live factory view, the emergency orchestrator, the incident report, and an outward-facing MCP server.

The Safety Council

The Council is the core of Corrix, and its defining property is deliberate information starvation. Five specialized agents evaluate every incident, and four of them are prevented, by construction, from seeing the whole picture:

AGENT	PERSPECTIVE	DATA ACCESS
Process Safety Engineer	Is the physical condition abnormal?	Sensor stream only
Permit Control Officer	Is authorized work compatible with conditions?	Permit and shift records only
Shift Operations	Is human-factors risk elevated right now?	Permit and shift records only
Site Safety Observer	Does site observation corroborate or contradict?	Computer vision and worker location only
Chair	What is the compound verdict?	The four agents' structured reports

Each agent's tool bindings are restricted to its own MCP server, and our test suite verifies the restriction directly by inspecting the bound tool lists, not by trusting a prompt. Only the Chair receives all four reports. This mirrors, deliberately, the same silo structure that caused the real anchor incident: the difference is that Corrix adds the one thing the real plant lacked, a synthesis layer that fuses the perspectives in time.

The claim this design earns: when Corrix says compound risk only emerges from fusion, that is a property of the architecture that can be checked in the code, not a line in a pitch.

Human in the loop, for real

Before the Chair rules, the state machine genuinely pauses for eight seconds. A safety officer can inject a note, for example that a permit was already suspended manually, and the note is written into the graph's checkpointed state; the Chair's verdict visibly incorporates it. This is a LangGraph interrupt, not a cosmetic progress bar, and it encodes a principle: Corrix is a decision-support system, never an autonomous controller. It notifies and documents. The final call always sits with a human.

Engineered for a live demo's worst moment

Inference runs on Groq first for sub-second latency, with automatic failover to Gemini. If both providers fail mid-demo, the system does not crash and does not fake confidence: it returns a clearly-labeled rule-based fallback verdict with zero stated confidence and a recommendation for manual review. We tested this by deliberately breaking both providers and watching the system stay honest.

Detection and foresight

Three independent ways to convene the Council

TRIGGER	HOW IT WORKS
Rule and threshold	Rolling z-score anomaly scoring per zone against a calibrated baseline, plus a deterministic permit-conflict rule table gated on hazard class. Fast, fully auditable, and the evaluation baseline.
Joint-evidence novelty	A Mahalanobis-distance model over the full joint evidence vector, fitted on normal-day runs. It escalates statistically unusual combinations that match no scripted pattern, and the verdict honestly says so rather than pretending to recognize a named pattern.
Memory retrieval	Similarity search against stored past misses. It catches the sub-threshold recurrence the other two paths would let pass, which is how the self-improving loop pays off operationally.

The novelty trigger answers the hardest question a judge can ask: what happens with a combination nobody wrote down, not the brief's named examples and not our own scenarios? Most systems built on a fixed scenario list quietly reduce to that list. Corrix demonstrates the opposite live, through an Open Challenge that draws a pre-validated but unrehearsed evidence combination and runs it through the novelty path in front of the audience.

From detection to action

Time-to-Critical

A Monte Carlo rollout reuses the simulator's own calibrated stepping function to project hundreds of stochastic futures, reported as a probability band: "68 percent chance of CRITICAL within 15 to 22 minutes."

Risk-aware evacuation

Dijkstra over the zone adjacency graph with edge weights inflated for elevated zones, so the route avoids other danger on the way to the nearest assembly point.

Spatial risk propagation

A decaying breadth-first search over the same graph, weighted by hazard class, estimating where the risk migrates if uncontained.

What-if mitigation

Isolating the source or adding ventilation re-runs the real forecaster with a reduced source term; procedural actions state honestly that they change exposure, not physics.

On a CRITICAL verdict, the Emergency Response Orchestrator fires a real notification carrying a timestamped, SHA-256-hashed evidence snapshot and the evacuation route, and a one-click incident report renders the full evidence trail to PDF. The first ten minutes of an incident become documented and coordinated instead of chaotic.

Regulatory intelligence and computer vision

One knowledge substrate, three questions

Corrix ingests real government source documents into a Neo4j graph with native vector search: the full text of the Factories Act 1948, a real OISD guideline document, and a verified DGMS circular, all chunked at clause boundaries with locally-computed embeddings. One substrate serves three query shapes through one interface:

- **Ask the regulation:** a retrieval-grounded chat that answers with checkable citations to real sections, never invented ones.
- **Has this happened before:** graph traversal over a near-miss corpus surfaces recurring patterns, the brief's Incident Pattern Intelligence direction.
- **Compliance check:** inspection-log entries are matched against live-retrieved requirements to flag deviations, the brief's Quality and Compliance Audit direction.

Every Council verdict additionally carries the specific clauses most relevant to its evidence, retrieved live. DGMS material is honestly labeled supplementary: DGMS statutorily governs mining, and pretending otherwise would be exactly the kind of forced citation this project refuses to make.

Computer vision, honestly scoped

Real

A YOLO11 model fine-tuned on the open Construction-PPE dataset runs genuine inference on site footage, detecting people and PPE classes: helmet, vest, gloves, boots, goggles, and their absence. Detections feed the Site Safety Observer agent's evidence.

Simulated, labeled as such

The zone and badge correlation on top of a detection, for example "person detected with no matching badge scan," is checked against the simulated worker-location stream. The split is encoded in the data schema itself as separate source fields, not just a UI label.

A measurable difference, not a decoration: the Site Safety Observer's assessment demonstrably changes when a vision event is present versus absent on otherwise identical evidence. The camera is an input to the reasoning, not a screensaver.

Data strategy and radical honesty

No public dataset of real Indian plant SCADA, gas-sensor, or permit-to-work data exists. This is proprietary, safety-sensitive information that no company publishes, and every team addressing this problem statement faces the identical constraint. What differentiates Corrix is a physics-informed, externally-validated simulation methodology, disclosed openly inside the product rather than hidden or apologized for.

Physics-informed simulation

Gas concentration follows a mean-reverting Ornstein-Uhlenbeck process representing normal ventilation equilibrium, with an injectable source term for scripted leak and accumulation events. Permit, shift, and worker-location generators produce structured records mirroring real system schemas, all deterministically seeded so every scenario is reproducible.

External validation: the simulator's noise-to-signal ratio, 0.043 to 0.055, falls inside the range measured from the real SWaT industrial control system dataset, 0.012 to 0.117. Our synthetic sensors were checked against how a real physical plant behaves, not tuned by eye.

What is real, what is simulated

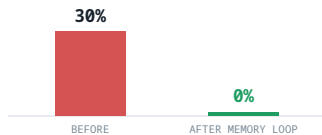
COMPONENT	STATUS
LLM reasoning: live Groq inference, Gemini failover	Real
Neo4j GraphRAG substrate and retrieval	Real
Regulatory source text: OISD, Factories Act 1948, DGMS	Real
Computer-vision inference: fine-tuned YOLO11 forward pass	Real
MCP integration, inward and outward	Real
Monte Carlo forecaster and evaluation methodology	Real
MQTT ingestion: broker, topics, parsing	Real protocol , software publishers
Gas and process sensor streams	Simulated, physics-informed, SWaT-validated
Permit-to-work and shift records	Simulated, mirroring real schemas
Worker location and badge pings	Simulated, zone-level granularity
Near-miss and inspection corpus	Mocked, labeled illustrative

This table is not buried in an appendix. It is presented inside the product, on a dedicated page that explains how every number on screen is produced. We treat the honesty as a feature: a judge evaluating a dozen synthetic-data projects in one sitting remembers the team that got the details right and said so.

Results

The evaluation harness is a subsystem of Corrix, not an afterthought. It scores the whole pipeline against the labeled scenario library with matched negative controls, and its results are committed with the code and shown inside the product. These are the headline numbers from real runs.

FALSE-NEGATIVE RATE, HELD OUT



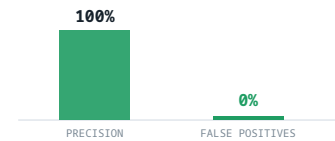
The metric that actually saves lives: missed incidents. Measured only on held-out scenarios the memory loop never trained on, so it is a generalization result, not memorization.

PREDICTION LEAD TIME



Minutes earlier than a legacy single-signal detector reaches the same escalation, before the incident threshold. Scenario S1 shows no gap and we disclose that rather than hide it.

DETECTION ACCURACY



On 20 matched negative-control runs the detector never cries wolf. The full pipeline's recall trade-off is disclosed in the in-product evaluation report, nothing smoothed over.

Regulatory coverage

OISD: 62 real guideline sections ingested.

Factories Act 1948: 113 real sections ingested.

DGMS: a verified circular, honestly labeled supplementary. Every verdict carries a live-retrieved citation; the corpus is checkable against the source PDFs.

Geospatial evidence

Every verdict carries zone-level worker occupancy from the badge-ping stream, a risk-aware evacuation route computed on the live zone-risk snapshot, and a propagation map over the plant adjacency graph on HIGH and CRITICAL.

Two further safeguards keep these numbers honest: negative controls are generated in matched volume to positive runs so the false-positive rate is not cherry-picked, and a confidence-calibration diagram compares the Council's stated confidence against its empirical accuracy.

Beyond the demo: a system that scales

Bring your own factory

The strongest evidence that Corrix is a platform rather than a rehearsed demonstration is that it has a second front door. Any facility can be onboarded through a wizard that captures its identity, zones with hazard classifications, an adjacency graph drawn in an interactive editor, workforce, shifts, and permit types. The graph the user draws is the exact input the evacuation router and propagation model consume.

Live data then flows in over MQTT, the same protocol real plants run, into a real broker, or through CSV historian replay, or from a built-in virtual sensor publisher. No hardware is required to prove the integration, and every simulated device is labeled as simulated. The Council, the detection triggers, the forecaster, and the router that serve the onboarded factory are the same objects the scripted demo calls. The data changes; the reasoning does not.

Integration is the same protocol, twice

Every internal data subsystem is a Model Context Protocol server that the Council consumes as a client, and Corrix's own compound-risk state is exposed back out as an MCP server. Any external MCP client can ask what the current compound risk in a zone is and why, and receive a live, grounded answer through the same interface the Council itself uses. In production terms this means integration into an existing plant stack is not a roadmap slide: the boundary already exists, on an open standard, in both directions.

The path to production

TODAY	IN A REAL DEPLOYMENT
Physics-informed simulated streams, SWaT-validated	Real SCADA and IoT feeds over the same MQTT boundary; layers 2 through 6 run unchanged
One camera path, fine-tuned PPE model	A multi-camera estate feeding the same observer agent
Eight-zone reference plant	The onboarding wizard already accepts arbitrary zone graphs
Free-tier inference with failover	The same failover pattern over production quotas

This is the honest scalability story: the part of the system that would change in production is precisely the part we already isolate behind a protocol boundary, and the part that is hard, the reasoning, the evaluation discipline, the explainability, is already built and measured.

Technology and status

LAYER	CHOICE	WHY
Agent orchestration	LangGraph	Native checkpointing and interrupts, which the Safety Officer Override requires structurally
Tool and data access	Model Context Protocol	One standard boundary for every data source, inward and outward
LLM inference	Groq primary, Gemini failover	Sub-second first tokens for a live convening, with automatic, tested failover
Knowledge substrate	Neo4j AuraDB, native vector search	One database serves both the knowledge graph and the retrieval corpus
Computer vision	Ultralytics YOLO11, fine-tuned	Real inference on real footage, trained on the open Construction-PPE dataset
Live ingestion	MQTT, WebSockets	The protocol real plants run, proven without requiring hardware
Backend	Python 3.13, FastAPI	Native WebSocket support, fast to build and test
Frontend	React 19, deck.gl, React Three Fiber	The command-center dashboard, 2D and 3D
Reports	Chromium HTML-to-PDF	Real incident documents with the evidence snapshot

Status

Corrix is built, tested, and deployed. The backend carries 415 passing tests covering the schemas, the simulation, every detection path, the Council and its silos, the regulatory substrate, the MCP servers, the evaluation harness, the memory loop, and the response pipeline. A public instance is live at corrix.duckdns.org.

In closing: the failure mode behind real incidents is data present but unacted upon. Corrix is the layer that acts on it: it fuses the five systems that were collectively blind at Visakhapatnam, reasons about their intersection the way a human review board would, explains itself with the regulation in hand, and provably gets better at catching the incidents it once missed. The signal is almost always there. Corrix is the layer that finally connects it.